

# MSc. in Computing Practicum Approval Form

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## Section 1: Student Details

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|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Project Title:     | How did the COVID-19 pandemic affect CO2 emissions in major European cities, and what long-term impacts might this have on future emissions trends?                |
| Student ID:        | A20403046, 43499                                                                                                                                                   |
| Student name:      | Colin Ekedigwe, Osebor Evbodaghe                                                                                                                                   |
| Student email      | <a href="mailto:colin.ekedigwe2@mail.dcu.ie">colin.ekedigwe2@mail.dcu.ie</a> ,<br><a href="mailto:osebor.evbodaghe2@mail.dcu.ie">osebor.evbodaghe2@mail.dcu.ie</a> |
| Chosen major:      | Artificial Intelligence, Data Analysis                                                                                                                             |
| Supervisor         | Michael Scriney                                                                                                                                                    |
| Date of Submission | 24/11/24                                                                                                                                                           |

## Section 2: About your Practicum

### What is the topic of your proposed practicum?

The topic for our practicum is focused on the impact of the COVID-19 pandemic on CO2 emissions in major European cities. We aim to analyze how lockdowns and changes in human activity influenced emissions. CO2 is a critical greenhouse gas contributing to climate change, and understanding its trends during the pandemic is vital. Previous studies have highlighted significant emissions reductions during lockdown periods, but long-term impacts remain unclear. By exploring these trends, we can identify structural changes and behaviors that could inform future urban sustainability policies and climate action strategies, helping cities maintain lower emissions post-pandemic.

### Please provide details of the papers you have read on this topic

1. Nicolini, G., et al. (2022) 'Direct observations of CO2 emission reductions due to COVID-19 lockdown across European urban districts', Science of The Total Environment, 830, 154662.

This paper analyzed CO2 fluxes measured by 13 eddy-covariance stations in 11 European cities before and during the pandemic, finding reductions of 5% to 87% across districts during initial lockdowns, with emissions quickly rebounding as restrictions were lifted.

The study focused on direct measurements in specific urban districts, not providing a comprehensive analysis of entire cities or countries.

2. Moran, D., et al., “Estimating CO2 emissions for 108,000 European cities,” *Earth System Science Data*, 14, 845–864. DOI: 10.5194/essd-14-845-2022.

This paper developed a comprehensive emissions inventory for European cities using a combination of bottom-up and top-down methods. Emphasized the need for accurate local inventories for climate action.

It did not focus specifically on the COVID-19 pandemic’s impact or the long-term trends in emissions. Our work will center on pandemic-related changes and future implications.

3. Le Quéré, C., et al., “Temporary reduction in daily global CO2 emissions during the COVID-19 pandemic,” *Nature Climate Change*, 10, 647–653. DOI: 10.1038/s41558-020-0797-x.

This paper quantified global CO2 emissions reductions during the pandemic, attributing the decrease to reduced transport and industrial activity.

It focused on global rather than European city-specific data and lacked insights into potential long-term trends. Your project will address these gaps for European cities.

4. 7. Liu, Z., et al. (2020) 'Near-real-time monitoring of global CO2 emissions reveals the effects of the COVID-19 pandemic', *Nature Communications*, 11, 5172.

This paper developed a near-real-time global CO2 emissions dataset, showing daily emissions changes across different sectors and countries during the pandemic.

While providing country-level data, the study did not focus on city-specific emissions or use direct urban measurements.

5. Forster, P.M., et al. (2020) 'Current and future global climate impacts resulting from COVID-19', *Nature Climate Change*, 10, 913-919.

This study analyzed global CO2 emissions changes during the pandemic using mobility data and emissions inventories, projecting future climate impacts under different recovery scenarios.

The study provided global estimates but lacked detailed analysis of specific European urban areas.

### **How does your proposal relate to existing work on this topic described in these papers? (200 words)**

Our proposal builds upon the existing research by addressing key gaps highlighted in the selected papers. Nicolini et al. (2022) focused on direct CO2 measurements in urban districts but did not provide a city-wide or national analysis, limiting the broader implications of their findings. Moran et al. (2022) offered a well-rounded emissions inventory for European cities but did not explore the specific impact of the COVID-19 pandemic or long-term trends, which my work will address. Le Quéré et al. (2020) quantified global CO2 reductions during the pandemic but lacked European city-specific data and insights into long-term effects, which I will focus on in my analysis. Similarly, Liu et al. (2020) provided a near-real-time global emissions dataset but did

not emphasize urban-specific data or direct city-level measurements, leaving a gap for city-centric studies. Finally, Forster et al. (2020) presented global projections of climate impacts during COVID-19 but did not explore urban emissions in Europe, nor the nuanced implications for urban climate policies.

## **What are the research questions that you will attempt to answer?**

### **1. How did the COVID-19 pandemic affect CO2 emissions in major European cities?**

- What were the changes in emissions during different phases of the pandemic, such as lockdowns and reopening periods?
- How did these changes vary by sector, such as transportation, energy consumption, and industrial activity?

### **2. Why did CO2 emissions change during the pandemic?**

- What were the driving factors behind these changes eg. shifts in energy use
- How did differences in urban planning, transport infrastructure, and energy sources contribute to variations across cities?

### **3. What are the long-term implications of the pandemic on CO2 emissions in European cities?**

- What trends can be predicted for future emissions based on the observed data?
- How can these insights inform urban planning and policy decisions to support sustainable recovery and emissions reduction goals?

## **What software and programming environment will you use?**

We will perform data cleaning using Python on Jupyter notebook

## **Data Characteristics**

**Geographic Scope:** Covers emissions data from major European countries and cities.

**Time Period:** Includes emissions data from years leading up to, during, and after the COVID-19 pandemic, allowing for a detailed temporal analysis.

**Sectors:** Emissions are categorized by sectors such as transportation, energy, industry, agriculture, and waste.

**Gases:** Primarily focuses on CO2 but also includes other greenhouse gases (CH4, N2O, etc.) for comprehensive analysis.

## **Data Cleaning**

Remove irrelevant data columns or entries not related to the cities or sectors of interest.

Handle missing data by excluding incomplete records if needed

## **Describe the dataset**

<https://www.eea.europa.eu/en/analysis/maps-and-charts/greenhouse-gases-viewer-data-viewers> - The EEA Greenhouse Gases Data Viewer offers data on CO2 and other greenhouse gas emissions, which are essential for monitoring progress toward climate goals. The dataset is derived from official submissions by European countries under international climate frameworks.

Dataset size and features - it spans through multiple years including pre and post-pandemic periods. Sectors such as Energy, transport, industry, agriculture, etc are among, CO2 gases are available along with the data for all EU member states and some other nations. The dataset is disaggregated by economic sectors, allowing deeper insights into specific contributors to emissions.

## **What experiments do you expect to run?**

Some of the experiments we expect to run are

### **Descriptive Analysis**

**Experiment:** Analyze pre-pandemic, pandemic, and post-pandemic CO2 emissions trends across European cities using time-series plots.

**Output:** Visualizations (e.g., line charts, heatmaps) showing emissions trends over time by sector and city. And summary statistics highlighting the percentage changes in emissions during different phases of the pandemic.

### **Sector Impact Analysis**

**Experiment:** Investigate how emissions from specific sectors (e.g., transport, energy, and industry) were affected during the pandemic.

**Output:** Bar graphs or sectoral breakdowns illustrating the contribution of each sector to emissions changes. And insights into which sectors were most impacted by lockdown measures and recovery patterns.

### **Comparative City-Level Analysis**

**Experiment:** Compare emissions trends between major European cities to identify variation in responses to the pandemic.

**Output:** City rankings based on the magnitude of emissions reductions. And geographic visualizations (e.g., maps with emissions data overlaid) to highlight spatial patterns.

## **List of models to be evaluated**

Time-Series Models